

Northwest Airlines Flight 255

Flight model: Douglas MD-82
Date: August 16, 1987

From: Detroit, USA
To: Phoenix, USA

The crew in this flight had completed 2 out of 5 total destination assigned for the day.

Problem

The weather is seen to be stormy. The rain starts before takeoff. Right before the taxiing, there is a last minute change in the run way due to the rain condition. Right before take off, the captain could not set the auto throttle as his computer is was not set in the take off mode. When the plane achieved the lift, the plane starts to roll side to side. The plane had not achieved the required lift making it wings to slam to a light pole and then crashes onto the ground.

Investigation

The witnesses say that when the plane lifted off, it couldn't climb and flew in a nose up position, Which indicate it didn't have enough power or speed to provide the lift. Else, there is a possibility that the high winds resisted the lift off. Several witnesses saw engine fire before the crash. The investigation finds that the engine operation was normal. The flame was the result of the plane hitting the light post. The weather at the time of crash is examined. Thunder storm can cause heavy threat to the pilots. On further investigation, it is found the the **flaps and slats** located at the back and front side of the wing is not extended. The flaps and slats must be retracted during takeoff to provide enough lift. The planes behind the flight, had seen the flaps and slats extended at the time of takeoff. But the investigation team claims that the cables connecting flaps and slats show that they were not extended. This evidence is supported by the FDR. On hearing the recording from the DVR, it is found that the last minute runway change along with the weather condition would have distracted the pilots which would have made the pilots failing to extend the flaps. On further investigation, the crew had forget to perform the taxiing checklist which include the preparation of flight for take off. Usually, an alarm should have been triggered when the flight is to takeoff with flaps retracted. The technicians hear stall warning popping up from one of the speaker when the warning should be triggered on both the speaker. This was found that there is the P40 circuit breaker which when enabled switches the warning such as stall making it sound different. This is done by few pilots at at times it acts as a nuisance for the pilots which is not advisable. It is found that the work load in the cockpit increased at disturbing weather conditions, making pilots fail to perform vital check lists.

Changes that were brought after the incident

The NTSB had asked the NASA to create a check list which would provide less work load to the pilots before take off. The checklist were often badly designed and when pilots are interrupted when performing the check list they often forget where they had left. This made airlines to train pilots to return to the top of the check list and redo the checklist again which increases the work load. This made checklist made into smaller chunks so they would have to redo small portion rather than a big one. Then the electronic check list was introduced.

Reference

1. A Fateful Takeoff: The Tragic Saga of Northwest Airlines Flight 255 — Mayday: Air Disaster
<https://www.youtube.com/watch?v=Jtrs8nfpuPM>